Norms: Definition and Examples

1 Definition: Norm

Let V be a vector space over the field of real numbers $\mathbb{R}$ (or complex numbers $\mathbb{C}$). A **norm** on V is a function

$$\|\cdot\| : V \to \mathbb{R}$$

that assigns a strictly non-negative real number to every vector in V, satisfying the following four axioms for all vectors $x, y \in V$ and all scalars $c \in \mathbb{R}$:

1. **Positive Definiteness:** $\|x\| \geq 0$, and $\|x\| = 0$ if and only if $x = 0$. *(The length of a vector is positive, and only the zero vector has zero length.)*
2. **Absolute Homogeneity (Scaling):** $\|cx\| = |c| \, \|x\|$. *(Scaling a vector by c scales its length by the absolute value of c.)*
3. **Subadditivity (Triangle Inequality):** $\|x+y\| \leq \|x\| + \|y\|$. *(The length of the sum of two vectors is at most the sum of their lengths.)*
4. **Non-negativity of the norm:** $\|x\| \geq 0$ *(This is technically implied by 1 and 2, but often listed explicitly).*

A vector space V equipped with a norm is called a **normed vector space**, denoted $(V, \|\cdot\|)$.

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2 The L^p Norms (The Family of Norms)

In finite-dimensional spaces like $\mathbb{R}^n$, the most common norms are the L^p norms. For a vector $x = (x_1, x_2, \ldots, x_n) \in \mathbb{R}^n$ and a real number $p \geq 1$, the L^p norm is defined as:

$$\|x\|_p = \left(\sum_{i=1}^{n} |x_i|^p \right)^{1/p}$$

2.1 Example 1: The L^∞ Norm (Maximum Norm)

The L^∞ norm is the limit of the L^p norm as $p \to \infty$. It is defined as the maximum absolute value of the components of the vector:

$$\|x\|_\infty = \max_{1 \leq i \leq n} |x_i|$$

- **Intuition:** This measures the "worst-case" or largest single component of a vector.
- **In Discrepancy Theory:** If you have a set system $\mathcal{S} = \{S_1, \ldots, S_m\}$ and a coloring χ, the discrepancy vector is $v = (\chi(S_1), \ldots, \chi(S_m))$. The L^∞ norm is $\max_i |\chi(S_i)|$. Minimizing this norm over all colorings χ gives the standard definition of discrepancy: $\min_\chi \max_i |\chi(S_i)|$.

2.2 Example 2: The L^2 Norm (Euclidean Norm)

The L^2 norm corresponds to $p = 2$. It is the standard "straight-line" distance in Euclidean space:

$$\|x\|_2 = \sqrt{\sum_{i=1}^{n} |x_i|^2} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

- **Intuition:** This measures the standard length of a vector.
- **In Discrepancy Theory:** The formula in your image is exactly the L^2 norm applied to the discrepancy vector.

$$D_2(\mathcal{S}) = \min_{\chi} \sqrt{\chi(S_1)^2 + \cdots + \chi(S_m)^2} = \min_{\chi} \big\|(\chi(S_1), \ldots, \chi(S_m))\big\|_2$$

  As your text notes, the L^2 norm is "friendlier to manipulate algebraically" because it is a smooth function (unlike the max function), allowing mathematicians to use techniques like linear algebra, Fourier analysis, and calculus to find optimal colorings.

2.3 Example 3: The L^1 Norm (Taxicab / Manhattan Norm)

The L^1 norm corresponds to $p = 1$. It is the sum of the absolute values of the components:

$$\|x\|_1 = \sum_{i=1}^{n} |x_i|$$

- **Intuition:** This is the distance a taxicab would drive in a grid-based city like Manhattan (moving only horizontally and vertically).

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3 A Concrete Numerical Example

Let $v = (3, -4, 0, 1)$ be a vector in $\mathbb{R}^4$.

- L^1 **Norm:** $\|v\|_1 = |3| + |-4| + |0| + |1| = 3 + 4 + 0 + 1 = 8$
- L^2 **Norm:** $\|v\|_2 = \sqrt{3^2 + (-4)^2 + 0^2 + 1^2} = \sqrt{9 + 16 + 0 + 1} = \sqrt{26} \approx 5.099$
- L^∞ **Norm:** $\|v\|_\infty = \max(|3|, |-4|, |0|, |1|) = 4$

Notice that for any vector, $\|v\|_\infty \leq \|v\|_2 \leq \|v\|_1$. This relationship is a fundamental result in normed spaces: all norms on a finite-dimensional vector space are equivalent (they bound each other up to a constant factor depending on the dimension).